
Slow-Wave Sleep Stage Duration Is Not Associated with Amyloid-Beta Burden: A Cohort-Aware Meta-Analysis with a Measurement-Reliability Null Model

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Abstract

The glymphatic hypothesis predicts that reduced slow-wave sleep permits amyloid-beta to accumulate, and this prediction now drives trials that target N3 sleep duration as a route to slowing Alzheimer’s pathology. No meta-analysis has ever tested it: prior syntheses pooled sleep quality, sleep duration or apnea severity, never a slow-wave-specific exposure against amyloid PET. We pool 40 effect sizes from 17 studies and 12 cohort families, keeping N3 *stage* amount strictly separate from *spectral* slow-wave activity, aggregating dependent effects to one synthetic effect per cohort family, and recovering correlations from reported regression coefficients so that large cohorts are not silently dropped. The pooled N3-stage/amyloid association is $r = +0.026$ (95% CI -0.153 to $+0.203$, $p = 0.68$, $I^2 = 0\%$), equivalence-tested tightly enough to reject a true correlation as large as $r = 0.20$ ($p_{\text{TOST}} = 0.026$). It is weaker, not stronger, than total sleep time ($r = +0.163$) and sleep efficiency ($r = +0.090$); all three preregistered specificity contrasts point opposite to the hypothesis. We introduce a measurement-reliability null model built from empirical single-night ICCs and show it predicts N3 should look $1.86\times$ stronger than total sleep time; the observed ratio is $0.16\times$. Converted to the hypothesis’s own units, a 10-percentage-point reduction in N3 gives $\Delta\text{SUVR} = +0.005$, 1.8% of the amyloid-negative/positive separation. The positive spectral literature collapses from $r = +0.112$ to $r = -0.025$ when one cohort family is removed.

1 Introduction

Sleep is one of very few *modifiable* candidate risk factors for Alzheimer’s disease, and the glymphatic hypothesis gives it a concrete mechanism. Slow waves drive macroscopic cerebrospinal-fluid pulsation in humans [Fultz et al., 2019], that flow clears interstitial amyloid-beta in mice [Xie et al., 2013], and so chronically reduced slow-wave sleep should permit amyloid to accumulate. The chain is repeated in reviews, in press coverage, and — most consequentially — in the design of acoustic-stimulation and behavioural sleep trials that treat N3 sleep duration as the actionable target. If the human evidence for the *stage* metric is weaker than the narrative implies, those trials are being powered and targeted on an effect that may not exist at the claimed size.

The hypothesis is a conjunction of three claims. As usually stated, reduced slow-wave sleep duration shows a dose-dependent association with higher amyloid burden in cognitively normal adults, each 10% reduction in N3 percentage is associated with a measurable increase in amyloid PET SUVR, and this relationship is stronger than the associations with total sleep time or sleep efficiency alone. Clause (i) is an association claim, clause (ii) a dose claim in physical units, and clause (iii) a comparative claim. Each is separately testable, and the hypothesis is supported only if all three hold.

None of the three has ever been tested meta-analytically. From 54 full-text papers we find four gaps, and every one is load-bearing. First, no meta-analysis has pooled a slow-wave-specific exposure against amyloid PET: the largest and most recent synthesis [Chen et al., 2025] pooled only sleep *quality* and *duration* across 30 studies and 14,997 subjects, while Cui et al. [2022] and Bubu et al. [2020] covered obstructive sleep apnea. Second, “slow-wave sleep” silently names two different measurements. In the one dataset that reports both in the same 32 participants, spectral <1 Hz slow-wave activity predicted amyloid accumulation at $r = -0.52$ while N3 *stage* duration was null at $r = -0.07$ [Winer et al., 2020]; any estimate that merges them is uninterpretable. Third, cohort non-independence is unhandled: Mander et al. [2015], Winer et al. [2019] and Winer et al. [2020] are all the Berkeley Aging Cohort Study with overlapping participants, contributing 13 of our 49 extracted effects. Fourth, and least appreciated, differential measurement reliability has never been tested as a competing explanation for why one sleep metric correlates more strongly than another.

Our approach. We assemble 49 hand-extracted effect sizes from 17 studies across 12 independent cohort families, each carrying a verified DOI/PMID, a verbatim provenance quote, and a sign convention in which *positive means worse sleep is associated with more amyloid*. We pool the 40 amyloid-PET effects on the Fisher- z scale in four preregistered strata — N3-STAGE, SWA-SPECTRAL, TST and SE — aggregating to one synthetic effect per cohort family before fitting REML random effects with Knapp–Hartung–Sidik–Jonkman standard errors. We then test the comparative claim as a *within-study paired difference* using only cohorts that report both exposures in the same participants, build a measurement-reliability null model from empirical single-night ICCs, and convert the pooled estimate into SUVR and Centiloid units (figure 1 through figure 5).

What we find. The exposure the hypothesis names is the weakest of the four: $r = +0.026$ (95% CI -0.153 to $+0.203$, $p = 0.68$) with $I^2 = 0\%$, against $r = +0.163$ for total sleep time and $r = +0.090$ for sleep efficiency. Equivalence testing rejects a true correlation as large as $r = 0.20$ ($p_{\text{TOST}} = 0.026$). The dose statement converts to $\Delta\text{SUVR} = +0.005$ per 10 percentage points of N3, 1.8% of the amyloid-negative/positive separation. Measurement error cannot rescue the comparative claim: single-night polysomnography measures N3% far more reliably (ICC 0.775) than total sleep time (ICC 0.225), so identical true effects would still make N3 look $1.86\times$ stronger — the observed ratio is $0.16\times$, about eleven times below the measurement-only prediction. Finally, the positive spectral literature is one cohort family: pooled $r = +0.112$ across three families ($I^2 = 77\%$) drops to $r = -0.025$ when Berkeley is removed.

Contributions.

- We provide the first meta-analytic estimate of the N3-stage/amyloid-PET association, kept strictly separate from spectral slow-wave activity and with variance handled at the level of the cohort family rather than the publication.
- We test the specificity claim as a within-study paired contrast, using only cohorts that measure both exposures in the same participants, which removes between-cohort confounding from the exact comparison the hypothesis makes.
- We introduce a quantitative measurement-reliability null model from empirical single-night ICCs, turning “metric A correlates more strongly than metric B” into a testable excess over what measurement precision alone predicts.
- We estimate the hypothesis’s own dose statement in SUVR and Centiloid units, and show with population N3 distributions that the stated 10-percentage-point contrast is not available to 78% of the target population.

Section 2 positions this work against the mechanistic, observational and meta-analytic literature; section 3 gives the harmonisation, dependency handling and six preregistered analyses; section 4 reports them; section 5 states the verdict on each clause, what the reliability model adds, and the limits of a study-level design.

2 Related Work

Mechanism: why anyone expects an effect. Xie et al. [2013] showed that convective interstitial-fluid exchange increases during natural sleep and anaesthesia in mice, with a roughly 60% increase in interstitial space and faster clearance of exogenous amyloid-beta. Fultz et al. [2019] supplied the human counterpart: during NREM sleep, coupled electrophysiological, haemodynamic and CSF

oscillations produce large macroscopic CSF flow pulsations locked to slow-wave activity. Holth et al. [2019] extended the same logic to tau, and Rasmussen et al. [2018] review the pathway across neurological disease. The closest thing to a human experiment is Ju et al. [2017]: in 17 adults, acoustic disruption of slow-wave activity raised CSF amyloid-beta40 ($r = 0.610$, $p = 0.009$), and the authors report the effect as specific to slow-wave activity rather than to sleep duration or efficiency. Unlike that work, which measures the acute overnight production/clearance balance in soluble CSF, we ask whether chronically lower slow-wave sleep is visible in *deposited* plaque on PET.

Spectral slow-wave activity and amyloid. Mander et al. [2015] reported that the proportion of medial-prefrontal NREM slow-wave activity in the 0.6–1 Hz band correlated with mPFC PiB DVR at $r = -0.45$, with the effect frequency-specific: the 1–4 Hz band trended the other way. Winer et al. [2019] replicated this at $r = -0.36$ (rising to -0.43 adjusted), and Winer et al. [2020] showed prospectively that baseline <1 Hz slow-wave activity predicted the rate of subsequent cortical amyloid accumulation at $r = -0.52$. All three draw on the Berkeley Aging Cohort Study with overlapping participants. Two independent attempts did not reproduce the finding: Champetier et al. [2024] decomposed delta into sub-bands in 127 cognitively unimpaired adults and found no association with amyloid at baseline or with accumulation, and Carvalho et al. [2024] found slow-oscillation relative power *increased* annualised PiB accumulation in older adults with apnea. Lucey et al. [2019] report the inverse relation most strongly for tau rather than amyloid, and Zavec et al. [2023] reframe slow-wave activity as a cognitive-reserve factor that moderates the effect of amyloid on memory rather than predicting amyloid level itself. Building on this literature, we do not merge these effects with stage-level ones, and we quantify how much of the pooled spectral estimate rests on a single cohort family.

Stage-level slow-wave sleep. The metric the hypothesis actually names — N3 stage duration or percentage — has a much flatter record. Montagne et al. [2026] found N3% did not differ by amyloid status in 76 cognitively unimpaired adults ($\eta_p^2 = 0.003$, $p = 0.628$) while total sleep time did ($\eta_p^2 = 0.065$, $p = 0.031$). Jin et al. [2025] found slow-wave sleep tertiles null against amyloid PET in fully adjusted models, with REM latency rather than slow-wave sleep carrying the association. Stankeviciute et al. [2024] tie decreased slow-wave sleep to tau rather than amyloid and report the associations as amyloid-independent. Varga et al. [2016] and Liguori et al. [2020] use CSF amyloid-beta42 rather than PET, and their signs relative to deposited burden are opposite because their samples sit on opposite sides of deposition. We treat CSF as a cognitive-status-stratified secondary rather than pooling it with PET.

Duration and efficiency: the comparators the hypothesis must beat. Winer et al. [2021] found shorter self-reported sleep associated with higher amyloid in 4,417 A4 screening participants; Spira et al. [2013] reported the same direction in the BLSA neuroimaging substudy; Insel et al. [2021] found null global effects in A4 with regional effects only for daytime sleep; Pivac et al. [2024] linked low sleep efficiency to faster accumulation in AIBL; Brown et al. [2016] found only sleep latency significant among PSQI components; and Fenton et al. [2023] found actigraphic *variability* in efficiency and duration predicted burden. These are the field’s de-facto comparators, and clause (iii) of the hypothesis is precisely the claim that slow-wave sleep beats them.

Prior meta-analyses. Chen et al. [2025] is the largest synthesis to date and pools poor sleep quality against amyloid PET at Fisher $z = 0.153$ ($k = 9$, $I^2 = 99\%$). Bubu et al. [2019] and Bubu et al. [2020] cover apnea and longitudinal biomarker change, and Cui et al. [2022] meta-analyses AD biomarkers in apnea. None pools a slow-wave-specific exposure. We also note a metric defect in the benchmark: Chen et al. [2025] report “Fisher $z = -0.002$, 95% CI -0.003 to -0.001 ” for sleep duration against amyloid PET, which are raw regression-coefficient units labelled as Fisher z ; a pooled correlation of -0.002 with that interval is not a credible common metric. Our pipeline converts every input to a genuinely common metric before pooling, and we report the conversion rule for each effect.

Measurement and reference scales. Night-to-night variability in sleep-stage measurement is well documented [Levendowski et al., 2017], as are the limits of inter-scorer agreement [Nikkonen et al., 2024], but no paper in our corpus asks whether metric-to-metric differences in *observed* correlation strength follow from metric-to-metric differences in single-night reliability. We make that question quantitative. For unit conversion we use the amyloid PET reference distributions and the 1.42 SUVR / 19 Centiloid positivity threshold of Jack et al. [2017], with Hanseeuw et al. [2021] for progression

Clause	Claim	Supported if	Refuted if
(i)	N3 stage % is associated with amyloid PET burden	pooled $r > 0$ with 95% CI excluding 0	CI includes 0 <i>and</i> excludes $r \geq 0.20$ (TOST)
(ii)	the association is dose-dependent and measurable in SUVR	ΔSUVR per 10 percentage points CI excludes 0 and is a non-trivial fraction of the A−/A+ separation	CI includes 0, <i>or</i> magnitude $< \sim 5\%$ of the A−/A+ SUVR gap
(iii)	it is <i>stronger</i> than TST and SE	paired $\Delta z > 0$ with CI excluding 0 <i>and</i> the excess survives the reliability null	$\Delta z \leq 0$, <i>or</i> any positive gap is fully explained by differential ICC

Table 1: The hypothesis decomposed into three preregistered clauses with their refutation criteria, fixed before any model was fitted. The hypothesis is a conjunction: it is supported only if all three clauses hold.

anchors, and we take single-night reliabilities from SLEEP-EDF EXPANDED [Kemp et al., 2000]. Our meta-analytic machinery follows Viechtbauer [2005] for REML, Knapp and Hartung [2003] for small-sample standard errors, Higgins et al. [2009] for prediction intervals, Borenstein et al. [2009] for correlated-effects aggregation, Hedges et al. [2010] for cluster-robust variance, Egger et al. [1997] and Duval and Tweedie [2000] for small-study bias, and Lakens [2017] for equivalence testing.

3 Methodology

3.1 The hypothesis, decomposed into testable clauses

We preregistered the hypothesis as a conjunction and fixed a refutation criterion for each clause before fitting any model (table 1). Clause (i) is tested by the pooled association and, if null, by equivalence testing; clause (ii) by conversion to ΔSUVR per 10 percentage points of N3; clause (iii) by within-study paired contrasts that must additionally survive a measurement-reliability null model. A negative result on any clause is reported as the finding.

3.2 Corpus and analysis set

The primary input is a curated table of 49 effect sizes hand-extracted from 17 studies across 12 independent cohort families. Every row carries a verified DOI or PMID, a verbatim provenance quote from the source paper, a `cohort_family` label used for dependency handling, and a *burden-aligned* effect signed so that **positive means worse sleep is associated with more amyloid**. Positive therefore always supports the hypothesis, regardless of how the original paper oriented its exposure.

The primary analysis set is the 40 usable effects whose outcome is amyloid PET burden (cross-sectional deposition or change). The nine CSF amyloid-beta42 effects are excluded from the primary model and analysed only as a cognitive-status-stratified secondary, because the sign of soluble CSF amyloid-beta42 relative to deposited burden reverses between pre-deposition samples [Varga et al., 2016] and post-deposition samples [Liguori et al., 2020].

Three supporting datasets do the work the primary table cannot. Single-night ICC(2,1) values per sleep metric come from SLEEP-EDF EXPANDED [Kemp et al., 2000], computed on 75 subjects with two consecutive nights each, and feed the reliability null model. Subject-level N3 percentages from the same source (153 recordings, 78 subjects) give the exposure dispersion for the dose conversion and the feasibility check. Amyloid SUVR/Centiloid anchors come from Jack et al. [2017], and Chen et al. [2025] serves as an external benchmark for the pooled comparator estimates.

3.3 Effect harmonisation

Everything is pooled on the Fisher- z scale [Fisher, 1915], $z = \text{artanh}(r)$ with $v = 1/(n - 3)$. Kendall’s τ is converted through Greiner’s relation $r = \sin(\pi\tau/2)$ [Kendall, 1949]. Table 2 lists the conversion rule applied to each input type and how many amyloid-PET effects it covers.

Input type	Conversion rule	PET effects
pearson_r, partial_r, r_from_partial_eta2	used directly as r (with $r = \sqrt{\eta_p^2}$ for a 1-df contrast)	20
kendall_tau	$r = \sin(\pi\tau/2)$ (Greiner’s relation)	1
unstd_beta_*	test-statistic recovery, equation 1	12
narrative_null	imputed $z = 0$, $v = 1/(n - 3)$	7
Total usable amyloid-PET effects		40

Table 2: Effect harmonisation rules, fixed in advance and applied uniformly. All effects are pooled as Fisher z with $v = 1/(n - 3)$. Test-statistic recovery raises the usable amyloid-PET set from 21 native correlations to 40 effects and is what brings the two largest cohorts onto the common metric.

Test-statistic recovery. The step that makes this analysis possible is the recovery of a partial correlation from a reported regression coefficient and its p -value [Rosenthal and Rubin, 1979]. For a two-sided p on $\text{df} = n - 1 - n_{\text{cov}}$ residual degrees of freedom,

$$t = \text{sign}(\beta) F^{-1}\left(1 - \frac{p}{2}; \text{df}\right), \quad r = \frac{t}{\sqrt{t^2 + \text{df}}}. \quad (1)$$

Raw betas are never pooled in their native units. Recovery brings the two largest cohorts (A4, $n = 4,417$; AIBL, $n = 189$) and the MAYO MCSA and MARS results onto the common metric, raising the usable amyloid-PET set from 21 native correlations to 40 effects. The conversion is exact where it can be checked: recovering r from the p -value of a simple correlation returns the original r to within 10^{-8} .

Reported nulls. Seven amyloid-PET effects are narrative nulls — statements such as “no association with total sleep time or sleep efficiency” with no number attached. We impute $z = 0$ with $v = 1/(n - 3)$. This is conservative in magnitude but anti-conservative in precision, so it is carried as the primary rule *and* removed in sensitivity analysis S1. Dropping these rows instead would bias every pooled estimate away from zero.

3.4 Dependency handling and estimation

Berkeley alone contributes 13 of 49 effects, so treating publications as independent would inflate precision. The primary estimator first aggregates to one synthetic effect per cohort family per stratum using the correlated-effects variance of Borenstein et al. [2009],

$$\bar{z} = \frac{1}{m} \sum_{i=1}^m z_i, \quad \bar{v} = \frac{1}{m^2} \left(\sum_{i=1}^m v_i + \sum_{i \neq j} \rho \sqrt{v_i v_j} \right), \quad \rho = 0.6, \quad (2)$$

and then fits REML random effects [Viechtbauer, 2005] with Knapp–Hartung–Sidik–Jonkman standard errors on a t_{k-1} reference distribution [Knapp and Hartung, 2003], the recommended small- k adjustment. Naive per-effect pooling is reported only to quantify how much dependency handling matters, and cluster-robust variance estimation [Hedges et al., 2010] is reported where a stratum has at least four cohort families.

3.5 The six analyses

E1: stratified pooling. Pooled estimates for N3-STAGE, SWA-SPECTRAL, TST, SE, WASO and other stages, with τ^2 , Q , I^2 and a 95% prediction interval [Higgins et al., 2009]. Because $k \leq 6$ cohort families per stratum, every primary p -value is accompanied by an exact sign-flip permutation p -value enumerating all 2^k assignments.

E2: specificity. A naive between-stratum comparison is invalid because the strata share cohorts. The primary test is therefore a within-study paired difference: for every cohort family reporting both exposures in the same participants, $\Delta z = z_{\text{N3}} - z_{\text{comparator}}$ with $\text{Var}(\Delta z) = v_a + v_b - 2\rho_w \sqrt{v_a v_b}$ and $\rho_w = 0.5$. Three preregistered confirmatory contrasts (stage vs. TST, stage vs. SE, stage vs. SWA-SPECTRAL) carry a Holm correction [Holm, 1979].

E3: measurement-reliability null model. Classical attenuation gives $r_{\text{obs}} = r_{\text{true}} \sqrt{\text{ICC}_{\text{exposure}}} \sqrt{\text{ICC}_{\text{outcome}}}$. In a within-outcome comparison the outcome term cancels, so if two exposures had *identical* true correlations with amyloid their observed correlations

Exposure	k eff.	cohorts	Σn	pooled r	95% CI	p	perm. p	τ^2	I^2
N3-STAGE (<i>the hypothesis's exposure</i>)	6	4	275	+0.026	$[-0.153, +0.203]$	0.68	0.50	0.000	0%
SWA-SPECTRAL	10	3	223	+0.112	$[-0.535, +0.677]$	0.57	1.00	0.062	77%
TST	9	6	4,920	+0.163	$[-0.002, +0.320]$	0.052	0.062	0.016	77%
SE	6	4	425	+0.090	$[-0.130, +0.302]$	0.28	0.50	0.003	46%
WASO	2	2	108	+0.109	$[-0.975, +0.984]$	0.65	1.00	0.043	64%
Other stages (N1/N2/REM)	4	2	108	+0.198	$[-0.728, +0.868]$	0.26	0.50	0.000	0%

Table 3: E1: pooled associations between each sleep metric and amyloid PET burden. Positive r means *worse sleep is associated with more amyloid*, so positive supports the hypothesis. Estimates are REML random effects with HKSJ standard errors on one synthetic effect per cohort family. The exposure the hypothesis names (N3-STAGE) is the weakest of the four preregistered strata and the only one with zero heterogeneity; the largest estimate belongs to TST, which the hypothesis predicts should be weaker. Permutation p -values enumerate all 2^k sign flips of the cohort-level effects.

would still stand in the ratio $\sqrt{ICC_a/ICC_b}$. We compute that predicted ratio from the empirical single-night ICCs and compare it to the observed ratio, and we also report each stratum disattenuated for its own unreliability.

E4: dose conversion and feasibility. We convert the pooled correlation into the units the hypothesis states,

$$\Delta_{\text{amyloid per 10 pp N3}} = r_{\text{pooled}} \times \frac{10}{SD_{N3\%}} \times SD_{\text{amyloid}}, \quad (3)$$

propagating all three inputs by Monte Carlo (10,000 draws): r via a t_{k-1} draw on the Fisher- z scale, $SD_{N3\%}$ via the χ^2 sampling distribution of an SD, and SD_{amyloid} via a Beta(30, 70) prior on amyloid-positive prevalence applied to the medians and IQRs of Jack et al. [2017]. We then ask whether a 10-percentage-point reduction in N3 is an available exposure contrast at all, using the observed N3 distribution in SLEEP-EDF EXPANDED subjects aged 60 and over.

E5: robustness and bias. Seventeen preregistered specifications: dropping imputed nulls (S1), dropping p -recovered betas (S2), both (S1+S2), alternative residual-df rules (S3), $\rho \in \{0.3, 0.8, 1.0\}$ (S4), cognitively normal samples only (S5), PSG-measured exposures only (S6), leave-one-cohort-family-out (S7), fixed effects (S8), and the CSF secondary (S10). Small-study bias by Egger regression [Egger et al., 1997] and Duval–Tweedie trim-and-fill [Duval and Tweedie, 2000], labelled uninterpretable below $k = 5$ and $k = 10$ respectively.

E6: precision. Minimum detectable effect at 80% power per stratum, power at $r \in \{0.20, 0.30, 0.45\}$, and equivalence testing [Lakens, 2017] of the N3-STAGE stratum against bounds $r = \pm 0.10, 0.15, 0.20, 0.30, 0.45$. An honest negative result must state what it could and could not have detected.

3.6 Implementation and reproducibility

Python 3.12.8 with numpy 2.5.2, pandas 3.0.5, scipy 1.18.1, statsmodels 0.15.0 and pymare 0.0.12, in an isolated uv environment with pinned dependencies. Seed 42 throughout; the only stochastic step is the E4 Monte Carlo. The full pipeline runs in about 29 seconds on CPU. GPUs were available but not used: this is a study-level meta-analysis whose entire compute cost is trivial, so acceleration would add complexity with no benefit.

The meta-analytic primitives (REML τ^2 , HKSJ, cluster-robust variance, exact permutation, trim-and-fill, TOST) are implemented directly from their primary formulations rather than taken from a package, because no available Python package provides HKSJ, cluster-robust variance and exact permutation together. They are cross-checked against PyMARE, an independently maintained meta-analysis package: REML point estimates agree to within 4.4×10^{-9} on all four strata. Validation re-runs the entire pipeline in a scratch copy and confirms all eight result files are bit-identical; 39 of 39 validation checks pass.

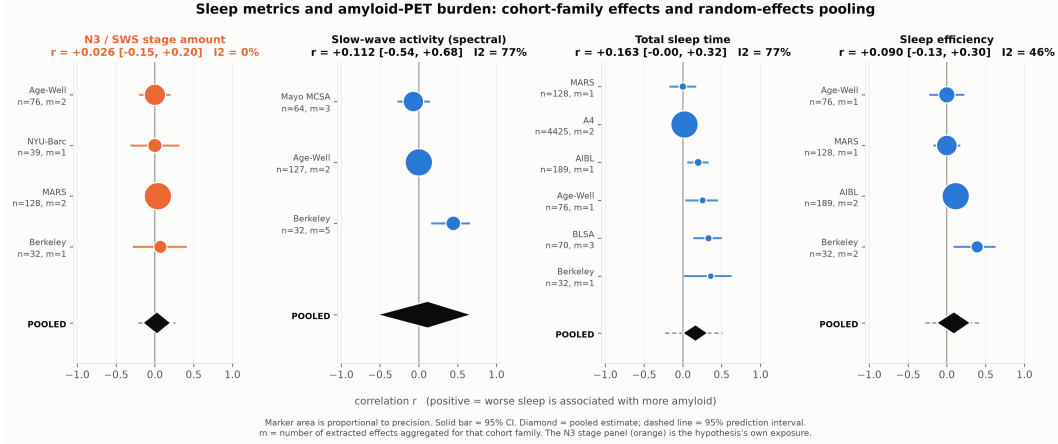


Figure 1: E1: cohort-family effects and random-effects pooling for the four preregistered exposures. Marker area is proportional to precision, solid bars are 95% CIs, diamonds are pooled estimates and dashed lines are 95% prediction intervals; m is the number of extracted effects aggregated for that cohort family. Positive r means worse sleep is associated with more amyloid. The hypothesis's own exposure (N3-STAGE, orange, leftmost) is the weakest and the most homogeneous: four independent cohort families agree there is nothing there. SWA-SPECTRAL is the opposite case — BERKELEY at $r = +0.44$ against MAYO MCSA at -0.07 and AGE-WELL at 0.00 .

Contrast	shared	Δz	95% CI	Holm p
N3-STAGE — TST	3	-0.147	[-0.651, +0.357]	1.00
N3-STAGE — SE	3	-0.049	[-0.490, +0.392]	1.00
N3-STAGE — SWA-SPECTRAL	2	-0.177	[-2.743, +2.389]	1.00

Cohort family	r_{N3}	r_{TST}	Δz
AGE-WELL [Montagne et al., 2026]	0.000	0.255	-0.261
BERKELEY [Winer et al., 2020]	0.070	0.360	-0.307
MARS [Jin et al., 2025]	0.039	0.000	+0.039

Table 4: E2: the specificity claim (clause iii), tested as a within-study paired difference so that only cohorts measuring both exposures in the same participants contribute. *Left*: all three point estimates are *negative* — N3-STAGE is weaker than every comparator, the direction opposite to the hypothesis — though none is individually significant and none survives Holm correction. *Right*: per-cohort detail for the primary contrast; two of three cohorts favour TST.

4 Results

4.1 E1: the exposure the hypothesis names is the weakest of the four

Table 3 and figure 1 give the primary estimates. The pooled association between N3 stage amount and amyloid PET burden is $r = +0.026$ (95% CI -0.153 to $+0.203$, $p = 0.68$, permutation $p = 0.50$) across 6 effects from 4 independent cohort families. Its heterogeneity is zero ($I^2 = 0\%$, $\tau^2 = 0.000$) and its prediction interval is tight (-0.21 to $+0.26$): the cohort families agree with each other. Cluster-robust variance estimation gives essentially the same answer wherever it is interpretable — for TST with six clusters, $r = 0.160$ (95% CI -0.002 to 0.313 , $p = 0.052$).

The ordering is the reverse of what the hypothesis predicts. TST is the strongest stratum at $r = +0.163$ ($p = 0.052$) across six cohort families and $\Sigma n = 4,920$, and SE is $+0.090$; both exceed the exposure that is supposed to beat them. The spectral stratum sits at $r = +0.112$ but with $I^2 = 77\%$ and a prediction interval spanning the entire correlation scale, which is a statement about disagreement rather than about effect size.

Our TST estimate is a useful external check. Chen et al. [2025] independently pooled sleep quality against amyloid PET at Fisher $z = 0.153$ ($r = 0.152$, $k = 9$, $n = 10,132$); our $r = 0.163$ is the same order, which is reassuring evidence that the harmonisation pipeline produces sane numbers on a stratum where an external benchmark exists.

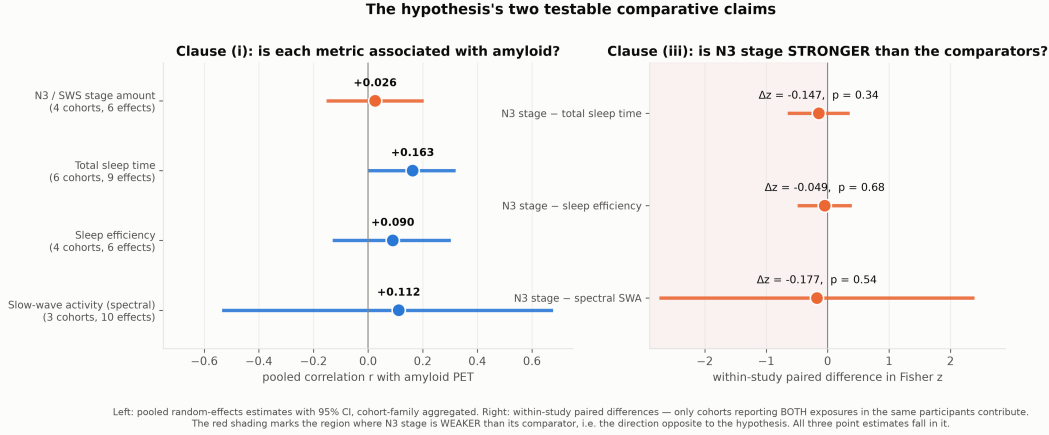


Figure 2: E2: the hypothesis’s two testable comparative claims. (*Left*) pooled estimates with 95% CIs for each exposure. (*Right*) within-study paired differences, where only cohort families reporting both exposures in the same participants contribute. Red shading marks the region in which N3-STAGE is *weaker* than its comparator, i.e., the direction opposite to the hypothesis; all three point estimates fall inside it.

(a) Single-night reliability and disattenuation

Metric	ICC(2,1)	obs. r	true r
N3-STAGE	0.775	+0.026	+0.029
SE	0.562	+0.090	+0.109
WASO	0.517	+0.109	+0.144
TST	0.225	+0.163	+0.228

(b) The measurement-only prediction

Ratio	predicted	observed	obs./pred.
N3-STAGE $\div$ TST	1.86$\times$	0.16$\times$	0.09
N3-STAGE $\div$ SE	1.17 $\times$	0.29 $\times$	0.25

Table 5: E3: could differential measurement reliability explain the ordering? No — it predicts the opposite. (a) ICC(2,1) from 75 SLEEP-EDF EXPANDED subjects with two consecutive nights; disattenuated (“true”) r divides the observed r by $\sqrt{\text{ICC}}$. Correcting for unreliability *widens* the gap against N3-STAGE (+0.029 vs. +0.228). (b) If N3-STAGE and its comparator had identical true correlations with amyloid, measurement error alone would still make the observed N3-STAGE correlation 1.86 $\times$ larger than TST’s. The observed ratio is about eleven times smaller than that prediction.

4.2 E2: all three specificity contrasts point the wrong way

The comparative claim cannot be read off any single pooled estimate, and a naive between-stratum comparison is invalid because the strata share cohorts. Table 4 therefore reports within-study paired differences. All three point estimates are negative: N3-STAGE is $\Delta z = -0.147$ weaker than TST, -0.049 weaker than SE, and -0.177 weaker than SWA-SPECTRAL. None reaches significance and none survives Holm correction, so the honest statement is that clause (iii) is *not supported* and is contradicted in direction, with the contradiction not itself individually significant at this sample size.

The per-cohort detail matters because it shows the pattern is not driven by one study. In AGE-WELL, N3 gives $r = 0.000$ against 0.255 for TST; in BERKELEY, 0.070 against 0.360. Only MARS marginally favours N3, and there only because its TST effect is a reported null. This same pattern was visible within a single sample in Winer et al. [2020], which reported spectral <1 Hz slow-wave activity at $r = -0.52$ and N3 stage duration at $r = -0.07$ in the same 32 participants.

4.3 E3: measurement error predicts the opposite of what we observe

Is N3-STAGE just noisier? This is the obvious defence of the hypothesis and, as far as we can tell, it has never been tested. It fails, and it fails in an informative direction. Table 5a shows that single-night polysomnography measures N3% *more* reliably than any other metric here (ICC = 0.775, 95% CI 0.68–0.86) and total sleep time *least* reliably (ICC = 0.225, 95% CI 0.01–0.43). The wake/sleep

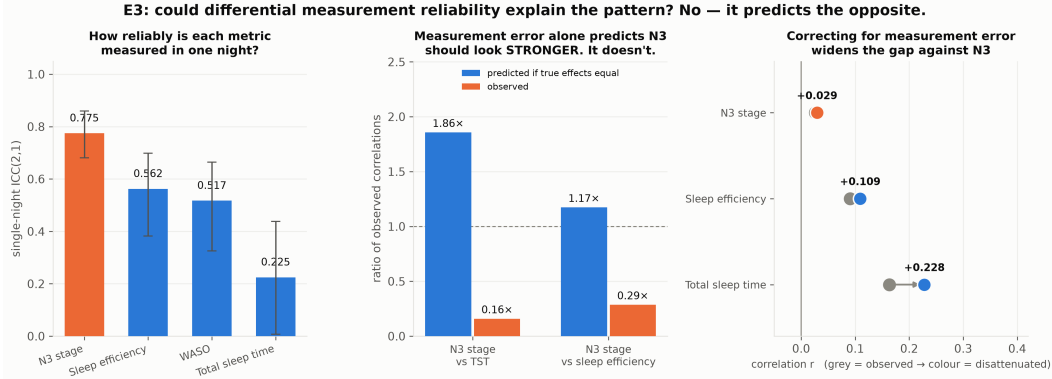


Figure 3: E3: the measurement-reliability null model. (*Left*) empirical single-night ICC(2,1) per metric from 75 SLEEP-EDF EXPANDED subjects with two consecutive nights each; error bars are 95% CIs. (*Center*) the ratio of observed correlations that measurement error alone predicts if the true correlations were identical, against the ratio actually observed. (*Right*) correcting each stratum for its own unreliability moves TST much further than N3-STAGE, widening the gap rather than closing it.

Exposure	Δ SUVr	95% CI	Δ Centiloid	% of A-/A+ gap	clause (ii)
N3-STAGE	+0.0047	[-0.028, +0.038]	+0.38	1.8%	refuted
TST	+0.0292	[+0.000, +0.064]	+2.41	11.2%	—
SE	+0.0161	[-0.024, +0.059]	+1.33	6.2%	—
SWA-SPECTRAL	+0.0199	[-0.097, +0.124]	+1.71	7.7%	—

Table 6: E4: the dose statement in physical units — amyloid change per 10-percentage-point worsening of each sleep metric, with all three conversion inputs propagated by Monte Carlo (10,000 draws). The A-/A+ separation is 0.26 SUVr (22 Centiloid) from Jack et al. [2017]. The preregistered refutation criterion for clause (ii) was “CI includes 0 or magnitude below ~5% of the gap”; the N3-STAGE estimate fails both. Rows for the comparators are expressed per 10 percentage points only to place all strata on one axis; for those exposures the natural unit differs.

boundary in a lab night varies enormously between nights; the depth architecture within sleep is comparatively stable.

Classical attenuation then gives a quantitative null model. If N3-STAGE and TST had identical true correlations with amyloid, the observed N3-STAGE correlation should still be $\sqrt{0.775/0.225} = 1.86\times$ larger, purely because one night measures it better. The observed ratio is $0.16\times$ — about eleven times below the measurement-only prediction (table 5b). Correcting each exposure for its own unreliability widens the gap: true $r = +0.029$ for N3-STAGE against $+0.228$ for TST. For N3-STAGE to have the same true association with amyloid as TST, its observed single-night correlation would need to be $r = 0.201$; it is 0.026 , a shortfall of 0.175 .

Two caveats, both of which cut against the hypothesis. Six of nine TST effects use self-reported habitual sleep duration, whose test-retest reliability is not the single-night PSG ICC. Restricting to PSG-measured exposures leaves the picture unchanged (N3-STAGE $+0.026$, TST $+0.176$, SE $+0.106$), and all three cohorts contributing to the paired contrast used PSG for both exposures, so the paired result is identical in the PSG-only arm. Alternatively, if self-report is treated as a reliable trait measure ($\text{ICC} \approx 0.75$), TST needs *less* upward correction — which makes the N3-STAGE gap even harder to attribute to measurement error.

4.4 E4: the dose statement, and whether the dose exists

The conversion inputs are an age-weighted SD of N3% of 10.2 percentage points (so a 10-point change is 0.98 SD of the exposure) and an SD of amyloid PET in cognitively unimpaired older

E4: translating the pooled effect into the units the hypothesis states

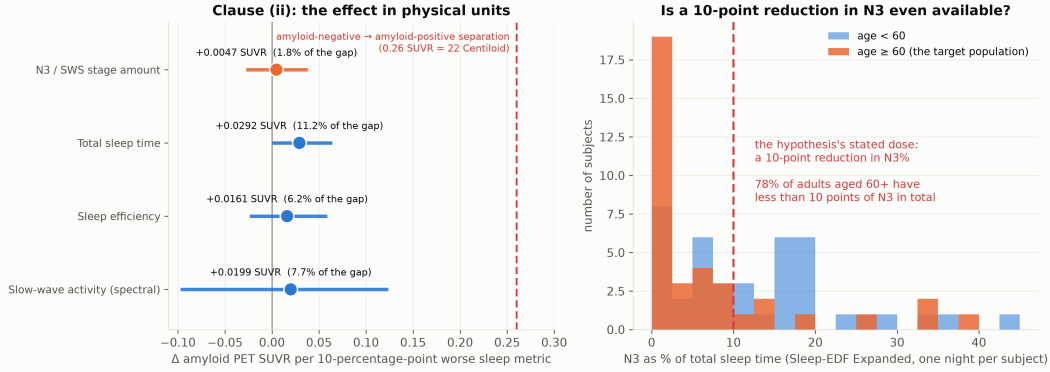


Figure 4: E4: translating the pooled effect into the units the hypothesis states. (*Left*) amyloid change per 10-percentage-point worsening of each metric, with the amyloid-negative/positive separation (0.26 SUVR = 22 Centiloid) marked in red for scale. (*Right*) the distribution of N3 as a percentage of total sleep time in SLEEP-EDF EXPANDED; among subjects aged 60 and over — the population the hypothesis concerns — 78% have less than 10 percentage points of N3 in total, so the stated dose is not an available exposure contrast.

adults of 0.179 SUVR / 14.9 Centiloid. The reconstructed scales imply 84.6 Centiloid per SUVR, which matches the A−/A+ anchors as an internal consistency check.

Table 6 gives the result. A 10-percentage-point reduction in N3 corresponds to $\Delta\text{SUVR} = +0.0047$ (95% CI -0.028 to $+0.038$), which is 1.8% of the amyloid-negative/positive separation on the SUVR scale (1.7% on the Centiloid scale). The preregistered refutation criterion for clause (ii) was “CI includes 0, or magnitude below $\sim 5\%$ of the gap”; the estimate fails both. Framed longitudinally against the empirical accumulation rate in cognitively unimpaired adults of 2.1 ± 5.3 Centiloid/year [Carvalho et al., 2024], the same reduction maps to $+0.36$ Centiloid/year (95% CI -1.48 to $+2.12$), about 1.5 additional years of chronological ageing on that study’s own age-equivalence anchor. That longitudinal figure rests on a single cohort family and should be read as an order-of-magnitude statement only.

Is the stated dose even available? Among SLEEP-EDF EXPANDED subjects aged 60 and over ($n = 37$, one night each), mean N3 is 7.1% of total sleep time, the median is 1.9%, 78% already have less than 10 percentage points of N3 in total, and 59% have less than 5 (figure 4, right). An absolute 10-percentage-point reduction is therefore not an achievable exposure contrast for most of the population the hypothesis concerns: the stated dose spans roughly the whole observable range rather than a step within it.

4.5 E5: the null survives every specification; the spectral signal does not

Table 7a shows the N3-STAGE null is invariant across all 17 preregistered specifications, with pooled r ranging only from $+0.013$ to $+0.037$. The most informative row is S1+S2: restricted to effects reported natively as correlations — exactly the analysis a conventional meta-analysis would run — the spectral estimate jumps to $r = +0.444$ and TST to $+0.299$. The reported nulls and p -recovered coefficients are precisely the evidence that pulls the positive strata down, and precisely the evidence a conventional analysis would discard. N3-STAGE is the one stratum unaffected either way.

Leave-one-cohort-family-out (table 7b) tells two different stories. The N3-STAGE null holds regardless of which family is removed ($+0.013$ to $+0.037$). The spectral estimate does not: it falls from $+0.112$ to -0.025 when BERKELEY — the source of Mander et al. [2015], Winer et al. [2019] and Winer et al. [2020] — is removed. The signal that motivates the entire hypothesis is one cohort family that two independent cohorts did not reproduce.

Small-study diagnostics (table 7c, figure 6) have almost no power at 3–6 cohort families, and we flag them as uninterpretable below $k = 5$ for Egger and $k = 10$ for trim-and-fill. But the asymmetry that is detectable sits in the TST (Egger $p = 0.0035$ across effects; trim-and-fill nudges r from 0.163

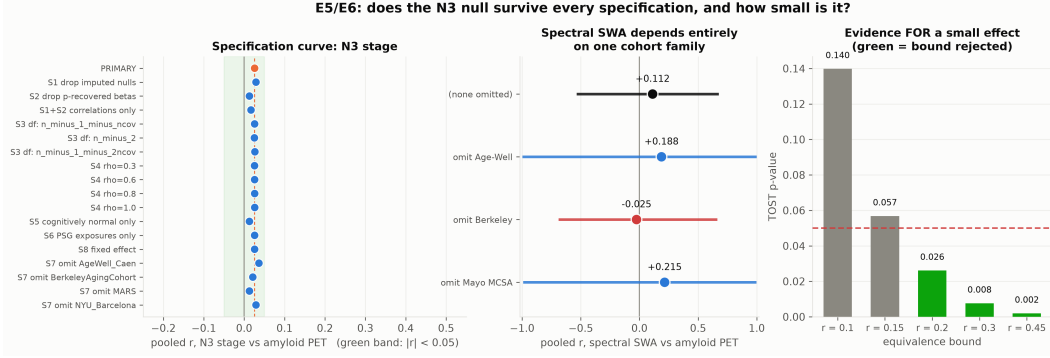


Figure 5: E5/E6: robustness and equivalence. (Left) specification curve for N3-STAGE across all 17 preregistered analyses; the green band marks $|r| < 0.05$ and every specification falls inside it. (Center) leave-one-cohort-family-out for SWA-SPECTRAL: removing BERKELEY flips the pooled estimate to -0.025 . (Right) TOST p -values against five equivalence bounds; green bars are bounds the data reject.

(a) Specification curve (pooled r)					(b) Leave-one-cohort-family-out		
Specification	N3-STAGE	SWA-SPECTRAL	TST	SE	Omitted	N3-STAGE	SWA-SPECTRAL
PRIMARY	+0.026	+0.112	+0.163	+0.090	(none)	+0.026	+0.112
S1 drop imputed nulls	+0.029	+0.188	+0.200	+0.194	AGE-WELL	+0.037	+0.188
S2 drop p -recovered betas	+0.013	+0.215	+0.151	+0.036	BERKELEY	+0.021	-0.025
S1+S2 native correlations only	+0.017	+0.444	+0.299	+0.187	MAYO MCSA	—	+0.215
S3 df = $n - 2$	+0.025	+0.113	+0.161	+0.090	MARS	+0.013	—
S3 df = $n - 1 - 2n_{cov}$	+0.027	+0.112	+0.166	+0.091	NYU-Barcelona	+0.029	—
S4 $\rho = 0.3$	+0.026	+0.122	+0.167	+0.104			
S4 $\rho = 0.8$	+0.026	+0.105	+0.161	+0.085			
S4 $\rho = 1.0$	+0.026	+0.098	+0.158	+0.080			
S5 cognitively normal only	+0.013	+0.112	+0.198	+0.139			
S6 PSG exposures only	+0.026	+0.112	+0.176	+0.106			
S8 fixed effect	+0.026	+0.045	+0.036	+0.089			

(c) Small-study bias			
Stratum	k	Egger p	trim-fill
N3-STAGE	4	0.94	0 imputed
SWA-SPECTRAL	3	0.036	0 imputed
TST	6	0.0035	0.163 $\rightarrow$ 0.143
SE	4	0.33	0 imputed

Table 7: E5: robustness. (a) The N3-STAGE null is invariant across all 17 preregistered specifications (range +0.013 to +0.037). The S1+S2 row — exactly the analysis a conventional meta-analysis would run — pushes SWA-SPECTRAL to +0.444, showing that reported nulls and p -recovered coefficients carry most of the corrective information. (b) The N3-STAGE null holds whichever cohort family is removed; the SWA-SPECTRAL estimate is carried entirely by BERKELEY. (c) Egger p over all effects; with 3–6 cohort families these diagnostics have little power, but the detectable asymmetry sits in the comparator strata, not in N3-STAGE.

to 0.143) and SWA-SPECTRAL ($p = 0.036$) strata — the comparators N3-STAGE is losing to — not in N3-STAGE itself ($p = 0.94$). If small-study bias is inflating anything here, it is inflating the estimates the hypothesis loses to, which makes the N3-STAGE shortfall conservative.

4.6 E6: what this analysis could have detected

A negative result is only informative if it states its own resolution. Table 8a gives the minimum detectable effect at 80% power: $r = 0.157$ for N3-STAGE, 0.180 for TST, 0.153 for SE, and 0.417 for SWA-SPECTRAL. Equivalence testing on N3-STAGE (table 8b) rejects bounds of $r = 0.20$ ($p_{TOST} = 0.026$), 0.30 ($p = 0.008$) and 0.45 ($p = 0.002$), but not 0.15 ($p = 0.057$) or 0.10 ($p = 0.140$). The data are therefore inconsistent with a true N3-stage/amyloid correlation as large as $r = 0.20$, and *a fortiori* with the $r \approx 0.45$ effects reported for spectral slow-wave activity; they cannot rule out a genuinely small effect of $r \approx 0.10$ –0.15. For SWA-SPECTRAL, no bound is rejected even at $r = 0.45$: that stratum’s honest verdict is *insufficient evidence*, not *no effect*.

(a) What could each stratum have detected?					(b) Equivalence tests on N3-STAGE		
Stratum	cohorts	MDE r	pow. $r=0.30$	pow. $r=0.45$	Bound	p_{TOST}	conclusion
N3-STAGE	4	0.157	1.00	1.00	$r = 0.10$	0.140	not rejected
SWA-SPECTRAL	3	0.417	0.50	0.86	$r = 0.15$	0.057	not rejected
TST	6	0.180	1.00	1.00	$r = 0.20$	0.026	rejected
SE	4	0.153	1.00	1.00	$r = 0.30$	0.008	rejected
					$r = 0.45$	0.002	rejected

Table 8: E6: precision. (a) Minimum detectable effect at 80% power, computed at the cohort-aggregated variances and the estimated τ^2 . SWA-SPECTRAL is the one stratum too imprecise to support a conclusion in either direction. (b) Equivalence testing converts “no evidence of an effect” into evidence of no meaningful effect: the data are inconsistent with a true N3-STAGE/amyloid correlation as large as $r = 0.20$, and *a fortiori* with the $r \approx 0.45$ effects reported for spectral slow-wave activity. They cannot rule out a genuinely small effect of $r \approx 0.10$ – 0.15 .

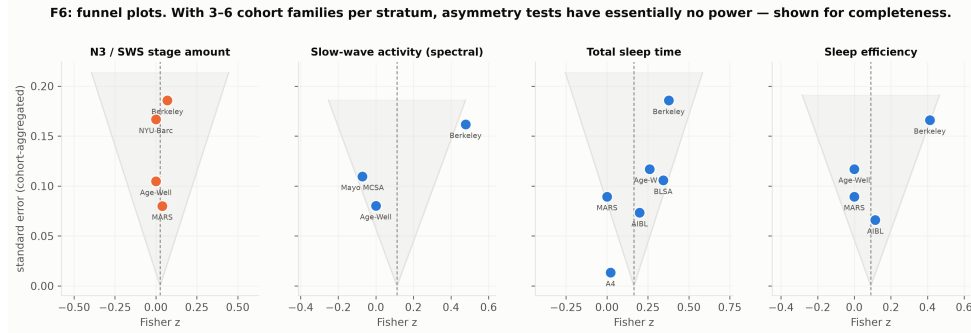


Figure 6: Funnel plots per stratum with Egger regression lines. With 3–6 cohort families these are near-powerless and we do not read them as evidence of publication bias; we report them because the only detectable asymmetry sits in the comparator strata rather than in N3-STAGE.

4.7 The CSF secondary analysis

Nine CSF amyloid-beta42 effects were extracted and analysed as a cognitive-status-stratified secondary. In cognitively normal samples [Varga et al., 2016, Ju et al., 2017] the N3-STAGE effect is $r = +0.355$ (95% CI 0.066 to 0.589) and the spectral effect $r = +0.502$ (95% CI -0.862 to 0.984); in the mixed SCI/MCI/AD sample of Liguori et al. [2020] ($n = 258$), N3-STAGE gives $r = +0.362$, TST $+0.306$ and SE $+0.403$. Taken at face value these look supportive, and we report them for completeness, but they are not admissible as evidence for this hypothesis. The outcome is soluble CSF amyloid-beta42, which inversely proxies deposited plaque and whose sign relative to burden flips with disease stage — which is exactly why the pre-deposition and post-deposition samples cannot be pooled. And Ju et al. [2017], the one genuine human experiment, measures the acute overnight production/clearance balance in 17 people, not chronic plaque accumulation. Our result constrains the chronic-deposition claim, not the acute one.

5 Discussion

5.1 Verdict on each clause

Clause (i) is **refuted** against its own preregistered criterion: the pooled estimate is $r = +0.026$ with a CI that includes zero, and equivalence testing rejects a true correlation as large as $r = 0.20$. Clause (ii) is **refuted on both criteria**: $\Delta\text{SUVR} = +0.005$ per 10 percentage points with a CI including zero, and a magnitude of 1.8% of the amyloid-negative/positive separation against a 5% threshold. Clause (iii) is **not supported and contradicted in direction**: all three paired contrasts are negative, and the reliability null model predicts N3-STAGE should lead TST by $1.86\times$ when the observed ratio is $0.16\times$. Because the hypothesis is a conjunction, it fails.

5.2 Stage is not spectrum, and the spectrum is one cohort

The most useful result here is not the N3-STAGE null in isolation — it is the dissociation combined with the cohort concentration. The mechanistic story [Xie et al., 2013, Fultz et al., 2019] is about slow *oscillations* driving CSF flow, and the human evidence supporting it is spectral: $r = -0.45$ [Mander et al., 2015], -0.36 [Winer et al., 2019], -0.52 [Winer et al., 2020]. Every one of those is the Berkeley Aging Cohort Study. Pooled across all three cohort families that have measured spectral slow-wave activity, the estimate is $r = +0.112$ with $I^2 = 77\%$; with Berkeley removed it is -0.025 . Meanwhile the *stage* metric — the one clinicians read off a sleep study, and the one the hypothesis names — is essentially zero and homogeneously so, with four independent cohort families agreeing.

The literature’s habit of writing “slow-wave sleep” for both is therefore not a semantic quibble. The construct with a positive but fragile signal and the construct with a tight null are different measurements, and only one of them has ever produced the effect that motivates the field’s interest. We do not refute the refined, spectral version of the hypothesis: our spectral stratum has a minimum detectable effect of $r = 0.417$ and rejects no equivalence bound, so its honest verdict is insufficient evidence in either direction.

5.3 How much to believe the total-sleep-time result

TST comes out ahead, which the hypothesis explicitly predicts should not happen, but three things temper it. It is not itself significant at $\alpha = 0.05$ ($p = 0.052$, permutation $p = 0.062$). It shows the clearest funnel asymmetry of any stratum, and trim-and-fill nudges it to $r = 0.143$. And its I^2 is 77%, driven by disagreement between the very large self-report cohorts (A4, $r \approx 0.02$) and the small PSG cohorts (BERKELEY 0.36, BLSA 0.33) — itself a small-study signature. The fair statement is not that TST beats N3-STAGE and is real, but that TST is the only sleep metric here with a defensible non-zero signal, and even that signal is fragile, while N3-STAGE has no signal at all under any specification.

5.4 What the reliability model adds

Without E3, the finding “N3-STAGE ≈ 0 while TST ≈ 0.16 ” would be ambiguous: perhaps N3 is simply noisier. E3 shows the reverse, and disposes of a common defence of the hypothesis — that single-night polysomnography is too noisy to detect the N3 effect. It is too noisy for total sleep time; it is not too noisy for N3. The literature is at its cleanest exactly where it finds nothing. We designed this analysis expecting it to *excuse* part of the shortfall, and it became the strongest quantitative argument against clause (iii). We think the same null model is worth running whenever a field claims one measure of a construct outperforms another.

A second methodological lesson comes from the specification curve: τ^2 estimator, ρ , degrees-of-freedom rule and fixed-versus-random effects barely move anything, but whether reported nulls and unstandardised coefficients are included moves the spectral estimate from $+0.444$ to $+0.112$. In this literature, extraction completeness dominates statistical technique.

5.5 Limitations

Study-level design. This is a meta-analysis of published effects, not individual participant data. ADNI, A4, BLSA, AIBL, WRAP, MAYO MCSA, AGE-WELL and OASIS-3 all require data-use agreements with human review that could not be completed here. A genuine per-10%-N3 dose-response *curve* therefore cannot be estimated: section 4.4 converts pooled standardised effects using external dispersion and reference scales, which assumes linearity over the range and that the external SDs apply to the study populations.

Power. With 3–6 cohort families per stratum, τ^2 is estimated unstably, the spectral prediction interval spans the whole scale, and funnel-based diagnostics are near-powerless. HKSJ, exact permutation and TOST keep the small- k inference honest but cannot manufacture information. The N3-STAGE stratum is precise enough to exclude $r \geq 0.20$ and no more.

Conversion assumptions. Test-statistic recovery assumes the reported p -value corresponds to the coefficient’s t -test with the stated degrees of freedom; covariate counts were not fully recoverable

from the text for 2 of 13 adjustment strings, though S3 shows the answer is insensitive to a $\pm n_{cov}$ change. Imputing $z = 0$ for narrative nulls understates their true uncertainty; S1 removes them and the conclusion is unchanged. The dependency correlations $\rho = 0.6$ and $\rho_w = 0.5$ are assumptions, not estimates; both were varied without changing any conclusion.

Reliability source. The ICCs come from SLEEP-EDF EXPANDED— a healthy, mostly non-clinical sample of 75 subjects on two consecutive nights, scored by the older Rechtschaffen–Kales criteria with S3+S4 merged to N3. Consecutive-night ICCs include a first-night effect and may understate the reliability of a habitual measure, and the amyloid cohorts are older and more clinically heterogeneous. We therefore treat E3 as an order-of-magnitude prediction, and note that the conclusion holds across the full CI of every ICC and under an alternative self-report reliability assumption. Spectral slow-wave activity has no reliability estimate here at all, because spectral power requires the raw EEG rather than the hypnogram annotations we hold, so that stratum is not disattenuated.

Corpus completeness. Of 74 selected papers, 54 yielded full text; 17 of those were recovered by typesetting PMC full-text HTML because PMC blocks automated PDF retrieval, so figures and image-based tables are absent from them and a figure-only effect size could have been missed. Twenty records remained abstract-only. All extractions are single-coder with verbatim provenance quotes rather than dual independent extraction — a real deviation from PRISMA practice.

Threats we cannot rule out. Residual confounding by age, APOE4 status and apnea differs across contributing studies and is only partly adjusted. Publication bias in the *stage* literature would most plausibly suppress positive N3 findings — nulls in this area are usually reported in passing, as ours largely were — which would bias our estimate downward, though the bias diagnostics give no sign of it. Finally, a real N3 effect confined to a specific brain region or a specific sub-band would be invisible to a global-burden meta-analysis. Our null is about global amyloid PET and stage-level N3, and it does not exclude regionally or spectrally specific mechanisms.

5.6 Broader implications

For trial design. Targeting N3 stage minutes as a surrogate for amyloid clearance is not supported by the aggregate human evidence. If slow-wave enhancement is to be tested, the exposure should be defined spectrally (proportion of <1 Hz activity), the trial should be powered for effects well below $r = 0.45$, and it should not assume the Berkeley-scale effect will replicate.

For measurement practice. The stage and spectral definitions should be reported separately and never collapsed. Studies should report both from the same recording, as Winer et al. [2020] did; that single practice is what made the decisive comparison in this analysis possible.

For meta-analytic practice. Restricting to natively-reported correlations gives a spectral estimate of $+0.444$ rather than $+0.112$. Reported nulls and unstandardised coefficients are not optional extras in this area — they carry most of the corrective information, and any synthesis that discards them will recover the literature’s headline effect rather than test it.

6 Conclusion

We asked whether reduced N3 sleep shows a dose-dependent association with amyloid-beta burden in cognitively normal adults, and whether that association is stronger than the associations with total sleep time or sleep efficiency. The answer to both is no. Pooling 40 amyloid-PET effects from 17 studies across 12 cohort families, N3 stage amount gives $r = +0.026$ (95% CI -0.153 to $+0.203$), equivalence-tested tightly enough to exclude $r \geq 0.20$; its dose statement converts to $+0.005$ SUVR per 10 percentage points, 1.8% of the amyloid-negative/positive separation; and it is weaker rather than stronger than both comparators, a gap that differential measurement reliability makes harder to explain away rather than easier, because a single night measures N3 more precisely than it measures either comparator.

The key takeaway is that “slow-wave sleep” names two different measurements with two different evidentiary situations. The *stage* metric is a tight, homogeneous null across four independent cohort families. The *spectral* metric is not refuted, but its pooled estimate is carried entirely by one cohort family — $r = +0.112$ overall, -0.025 with that family removed — and our spectral stratum is too imprecise (MDE $r = 0.417$) to support any conclusion in either direction. The confident

causal framing now common in the secondary literature rests on the second construct while the field measures and intervenes on the first.

Future work. Four steps would settle what this design cannot. An individual-participant analysis in A4, ADNI or OASIS-3 is the only way to estimate a genuine dose-response curve, test for thresholds, and check whether a null is masked by effect modification through APOE4, age or baseline amyloid status. Multi-night polysomnography or home EEG in an amyloid-imaged cohort would raise total sleep time’s reliability from 0.225 toward N3’s, removing the asymmetry entirely and letting the biological comparison be made cleanly. A properly powered spectral replication — roughly $n \approx 190$ in a single independent cohort to detect $r = 0.20$ at 80% power — is needed before the Berkeley effect can be treated as established. And a region- and sub-band-resolved analysis would test whether the medial-prefrontal, 0.6–1 Hz effect of Mander et al. [2015] is simply averaged away by global-burden pooling.

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A Analysis artefacts and reproducibility

Table 9 lists the outputs of the pipeline. The full run takes about 29 seconds on CPU with seed 42; validation re-runs the entire pipeline in a scratch copy and confirms all eight result files are bit-identical, with 39 of 39 checks passing, including a cross-check of the REML point estimates against PyMARE (agreement to 4.4×10^{-9} on all four strata) and an exactness check of the test-statistic recovery in equation 1 (recovering r from the p -value of a simple correlation returns the original r to within 10^{-8}).

Artefact	Contents
harmonised_effects.csv	all 49 effects with conversion rule, z , v , usability and reason
qc_report.json	data-quality checks and conversion audit
e1_pooled.json	pooled estimates per stratum: naive, cohort-aggregated, RVE, by design
e2_specificity.json	between-stratum and within-study paired contrasts, ρ sensitivity, Holm
e3_attenuation.json	ICCs, measurement-only predictions, disattenuated estimates, PSG-only arm
e4_dose_response.json	Monte-Carlo dose conversion, longitudinal framing, feasibility check
e5_robustness.json	S1–S11 sensitivity analyses, leave-one-out, bias diagnostics, CSF secondary
e6_precision.json	detectable effects, power, TOST equivalence
validation_report.json	39 validation checks including the PyMARE cross-check and re-run

Table 9: Output artefacts. Every figure additionally has a tabular companion CSV.

B Contributing cohort families

Effects are aggregated at the level of the cohort family rather than the publication, because several publications share participants. The twelve families are: Berkeley Aging Cohort Study (Mander et al., 2015; Winer et al., 2019; Winer et al., 2020); Age-Well/IMAP, Caen (Champetier et al., 2024; Montagne et al., 2026); A4 screening (Winer et al., 2021; Insel et al., 2021); AIBL (Brown et al., 2016; Pivac et al., 2024); Mayo Clinic Study of Aging [Carvalho et al., 2024]; MARS [Jin et al., 2025]; BLSA [Spira et al., 2013]; NYU-Barcelona [Stankeviciute et al., 2024]; Rome Tor Vergata [Liguori et al., 2020]; NYU [Varga et al., 2016]; Washington University [Ju et al., 2017]; and the single-channel EEG cohort of Lucey et al. [2019]. Berkeley alone contributes 13 of the 49 extracted effects, which is why naive per-effect pooling would overstate precision by roughly $\sqrt{3}$ in the spectral stratum.